

6 DNA replication occurs in interphase during the cell cycle.

(a) DNA ligase is involved in DNA replication.

Figure 6.1 shows a diagram of DNA ligase.

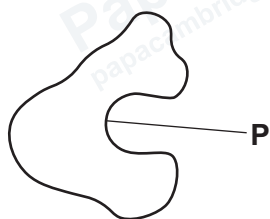


Figure 6.1

The site labelled **P** in Figure 6.1 is where the substrate binds.

With reference to the mode of action of enzymes, explain how DNA ligase allows successful DNA replication to occur.

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(b) State the phases that occur in interphase **and** outline the events that occur in each phase.

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(c) Mitosis occurs after interphase in the cell cycle.

Figure 6.2 shows plant cells in stages of mitosis.

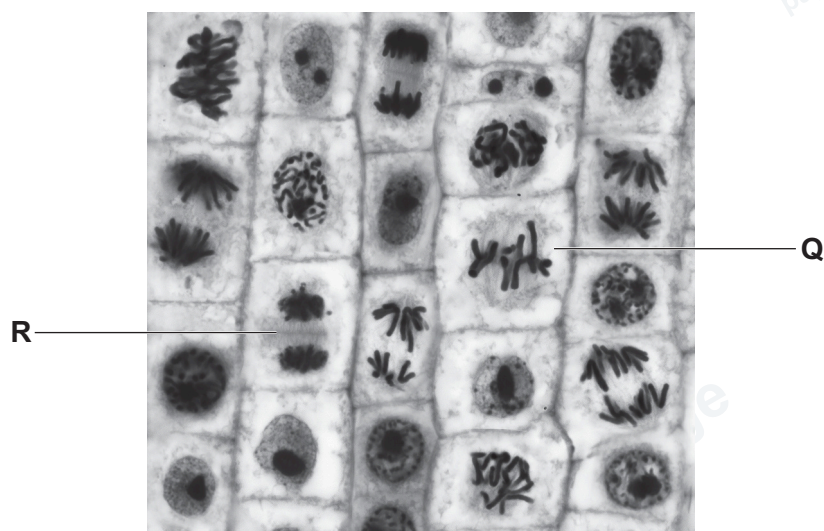


Figure 6.2

(i) Name the stage of mitosis that is occurring in the cell labelled **Q** in Figure 6.2.

..... [1]

(ii) State the process that is happening at the region labelled **R** in Figure 6.2.

..... [1]

[Total: 9]